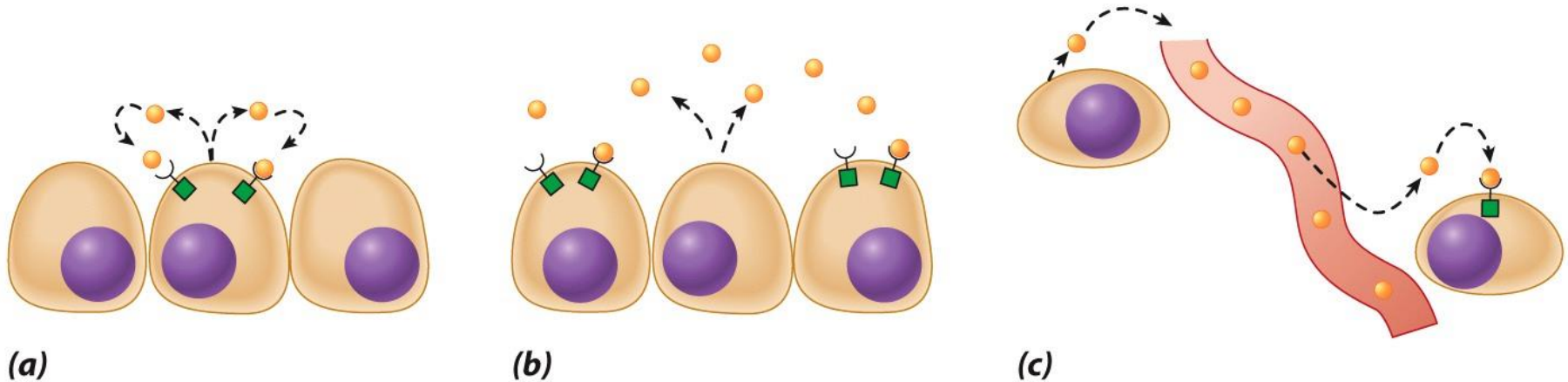


Cell Signaling Pathways



Jin-Wu Tsai (蔡金吾), Ph.D.
Professor
Institute of Brain Science
National Yang Ming Chiao Tung University

15.1 | The Basic Elements of Cell Signaling Systems



Copyright © John Wiley & Sons, Inc. All rights reserved.

Extracellular messenger molecules transmit messages between cells.

In autocrine signaling, the cell has receptors on its surface that respond to the messenger.

During paracrine signaling, messenger molecules travel short distances through extracellular space.

During endocrine signaling, messenger molecules reach their target cells through the bloodstream.

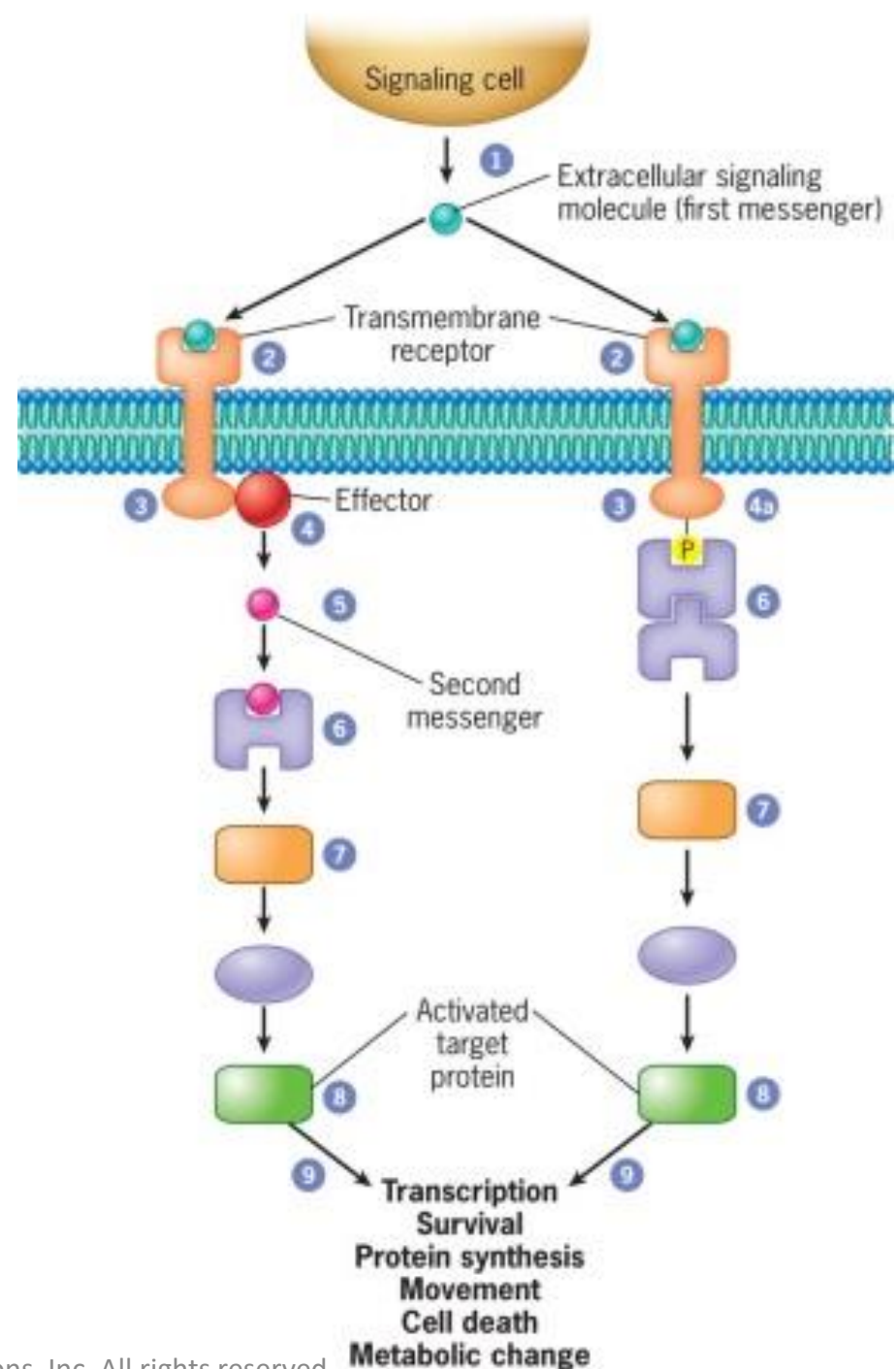
15.1 | The Basic Elements of Cell Signaling Systems

Receptors on or in target cells receive the message.

Some cell surface receptors generate an intracellular second messenger through an enzyme called an effector.

Second messengers are small substances that activate (or inactivate) specific proteins.

Other surface receptors recruit proteins to their intracellular domains at the plasma membrane.



15.1 | The Basic Elements of Cell Signaling Systems

Signaling pathways consist of a series of proteins.

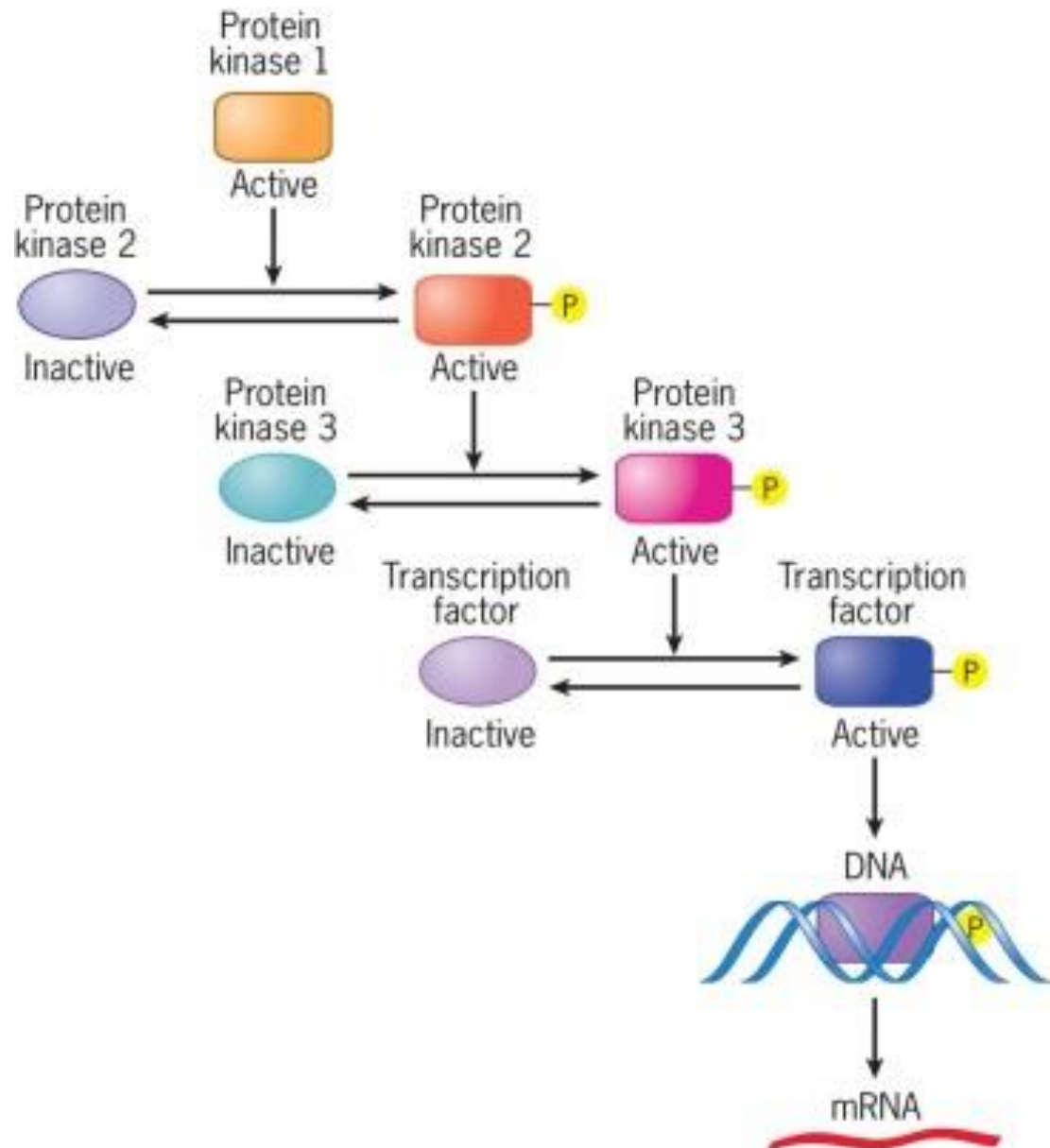
Each protein in a pathway alters the conformation of the next protein.

Protein conformation is usually altered by phosphorylation.

Kinases add phosphate groups while phosphatases remove them.

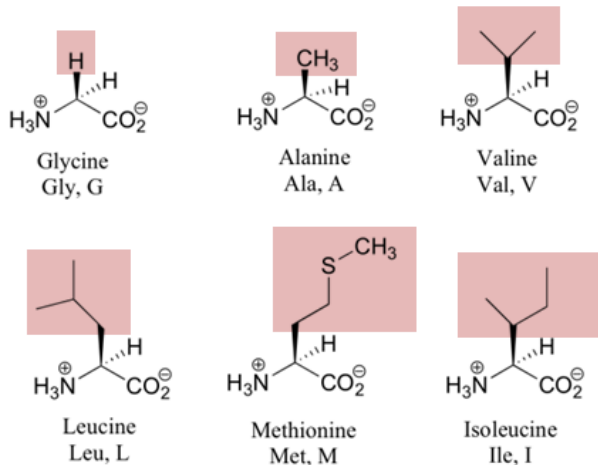
Target proteins ultimately receive a message to alter cell activity.

This overall process is called **signal transduction**.

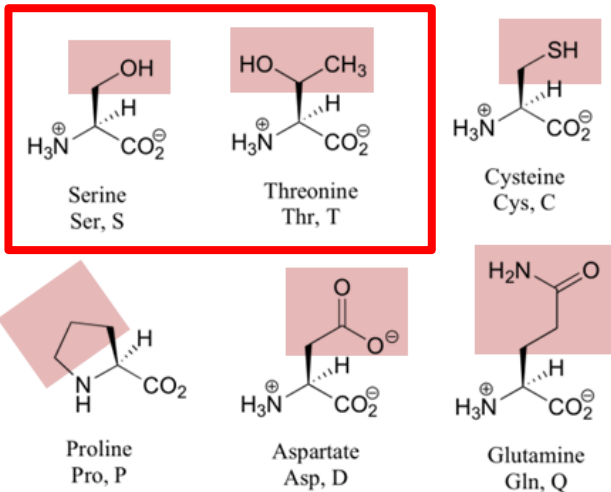


Amino acids

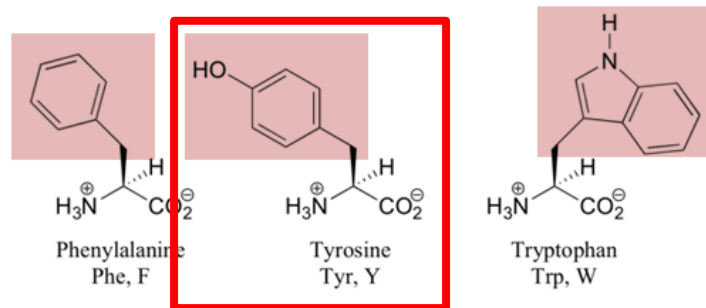
Nonpolar, aliphatic side groups



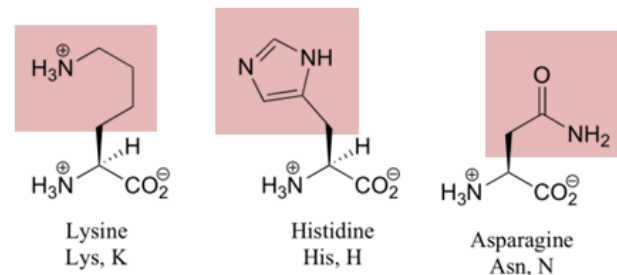
Polar, uncharged side groups



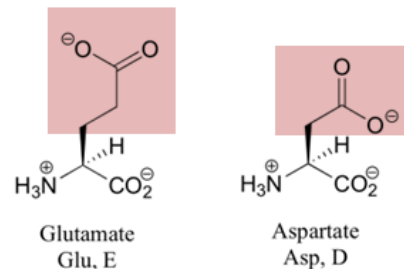
Aromatic side groups



Positively charged side groups



Negatively charged side groups



15.2 | A Survey of Extracellular Messengers and Their Receptors

Extracellular messengers include:

Small molecules such as amino acids and their derivatives (e.g. acetylcholine, epinephrine, dopamine).

Gases such as NO and CO.

Steroids, derived from cholesterol, regulate sexual differentiation, pregnancy, carbohydrate metabolism.

類固醇

Eicosanoids, derived from arachidonic acid, regulate processes including pain, inflammation, blood pressure, and blood clotting.

花生酸類

Peptides and proteins, including transmembrane proteins on the surface of an interacting cell, extracellular matrix interacting proteins, and secreted proteins.

15.2 | A Survey of Extracellular Messengers and Their Receptors

Receptor types include:

G-protein coupled receptors (GPCRs), which contain seven transmembrane α helices and activate GTP-binding proteins.

Receptor protein-tyrosine kinases (RTKs), which dimerize and activate their cytoplasmic protein-kinase domain to phosphorylate specific tyrosine residues of cytoplasmic substrate proteins.

Ligand gated channels, which conduct a flow of ions across the plasma membrane to change its potential.

Steroid hormone receptors, which function as ligand-regulated transcription factors.

Specific receptors such as B-and T-cell receptors which are involved in the response to foreign antigens.

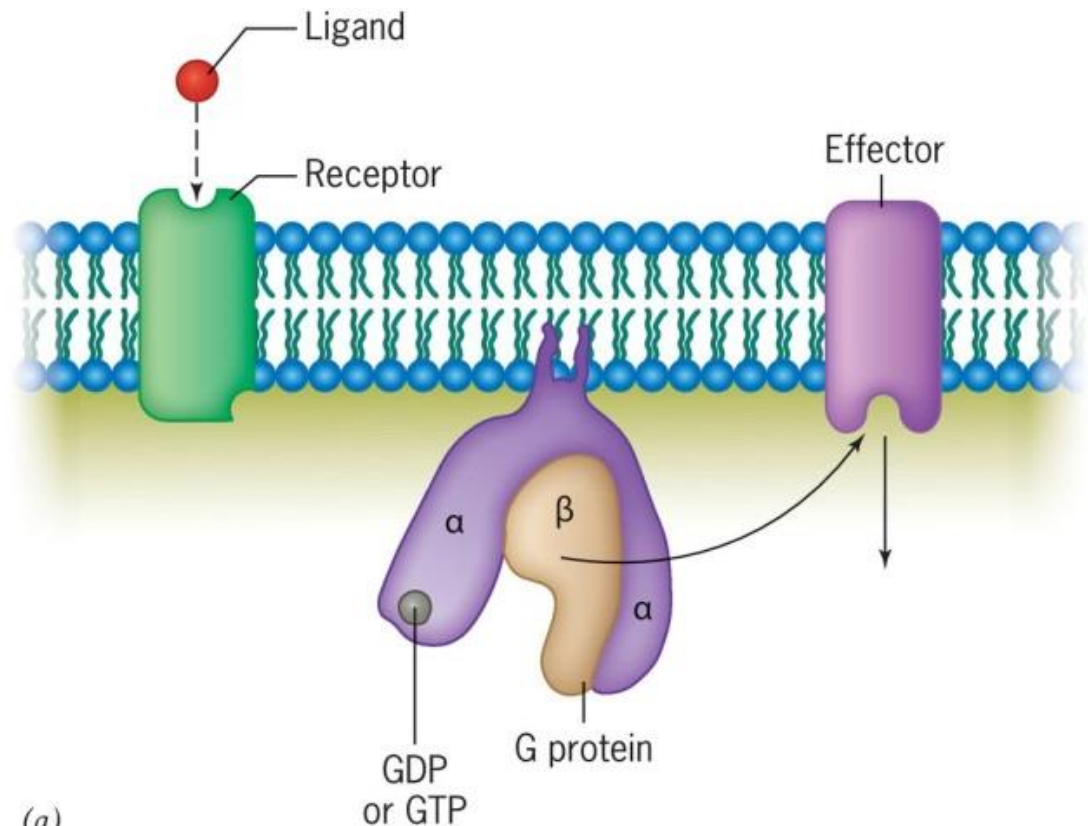
15.3 | Signal Transduction by G Protein-Coupled Receptors

G protein-coupled receptors (GPCRs)

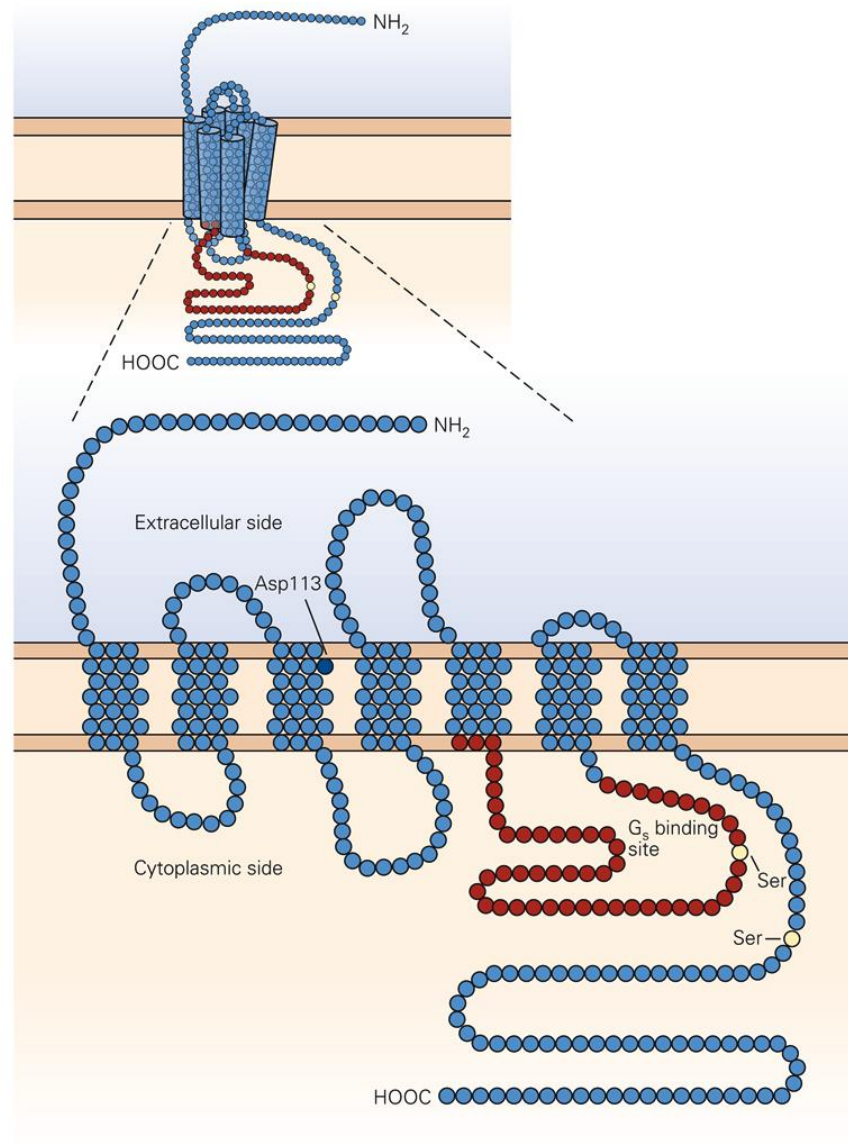
constitute the single largest superfamily of proteins encoded by animal genomes (1000s).

GPCRs have seven α -helical transmembrane domains and interact with G proteins.

Natural ligands: hormones (both plant and animal), neurotransmitters, opium derivatives, chemoattractants (odorants, tastants, and photons).

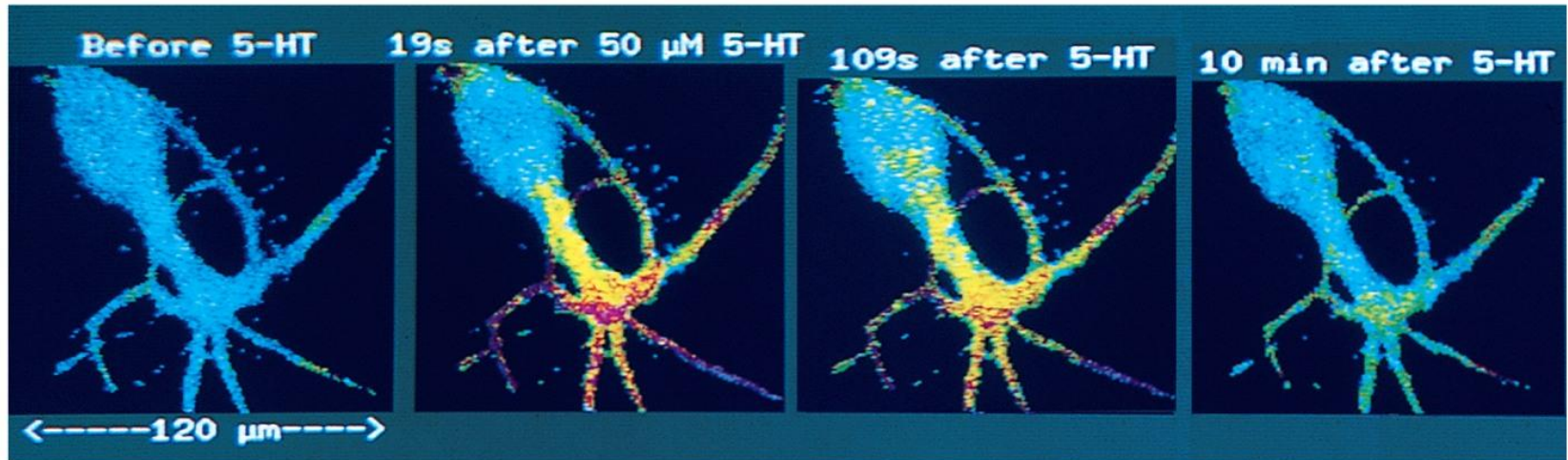


G protein-coupled receptors contain seven membrane-spanning domains



β_2 -adrenergic receptor

15.4 | Second Messengers



From Brian J. Bacskaï, et al., Science 260:223, 1993. Reprinted with permission from AAAS.

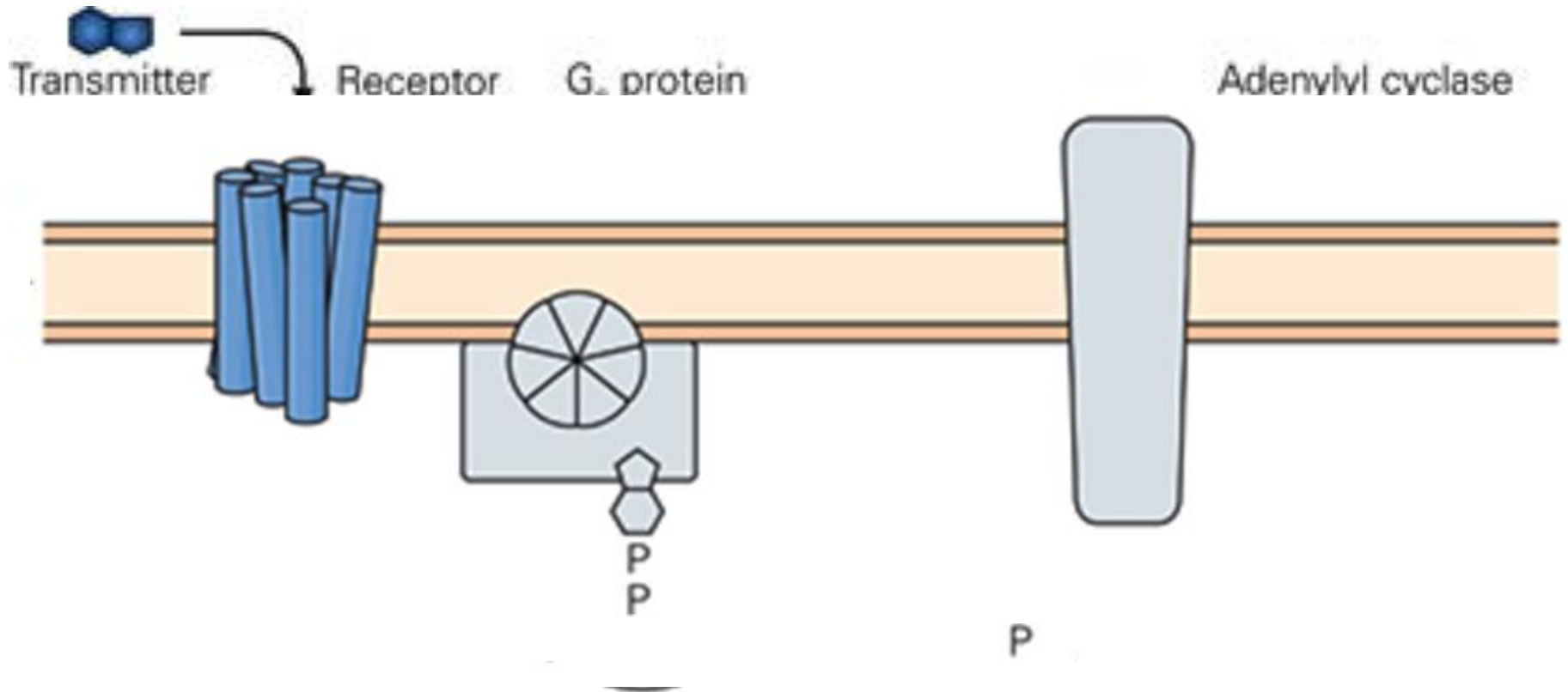
Cyclic AMP is a second messenger that diffuses to other sites in the cell.

The synthesis of cyclic AMP follows the binding of a first messenger, a hormone or other ligand, to a receptor at the outer surface of the cell.

Second messengers enable cells to mount a large-scale, coordinated response following stimulation by a single extracellular ligand.

Other second messengers include Ca²⁺, phosphoinositides, inositol trisphosphate, diacylglycerol, cGMP, and nitric oxide.

The cyclic AMP cycle



The cyclic AMP cycle

Transmitter binding alters conformation of receptor, exposing binding site for G_s protein.

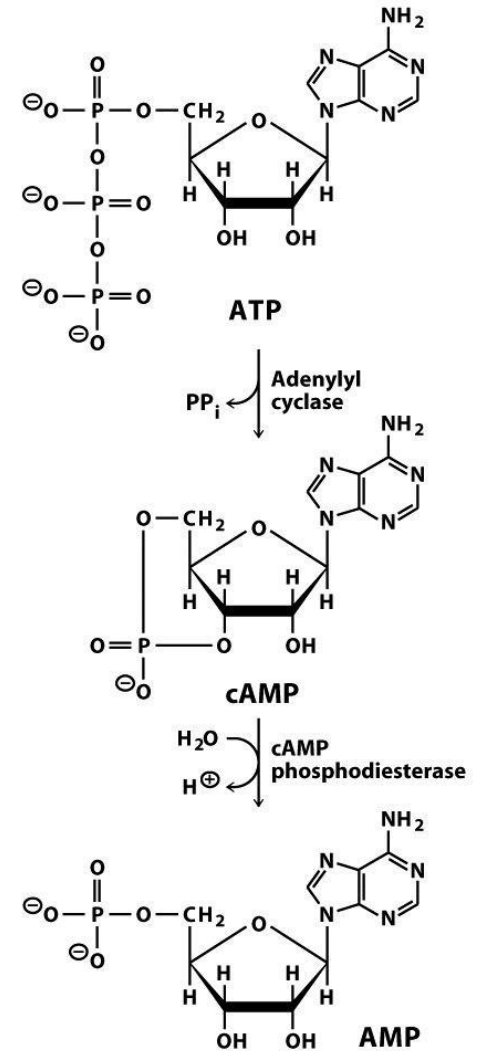
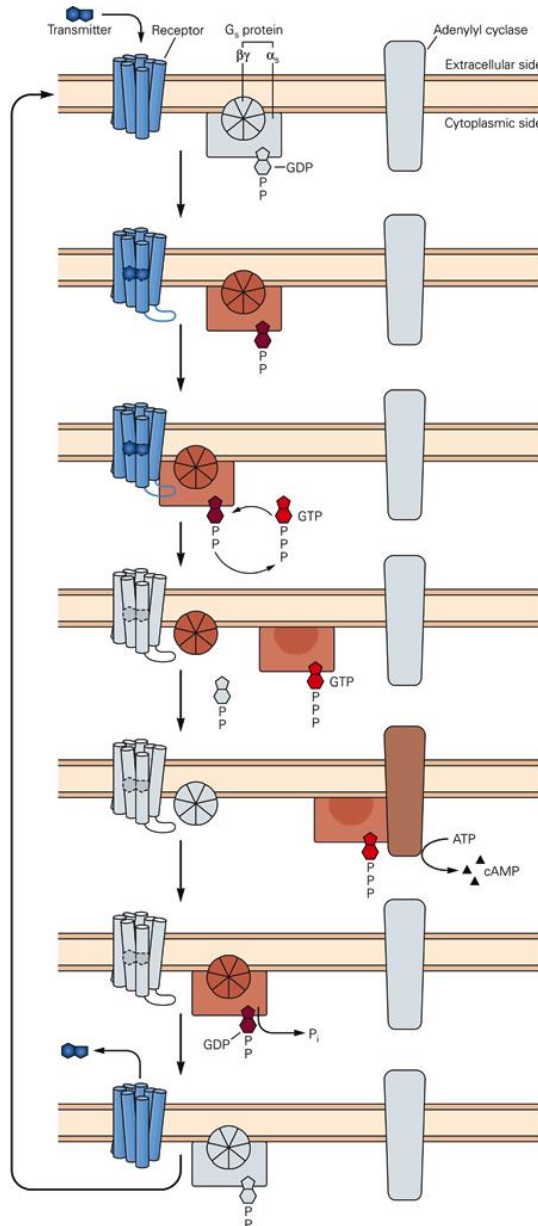
Diffusion in the bilayer leads to association of transmitter receptor complex with G_s protein, thereby activating it for GTP-GDP exchange.

Displacement of GDP by GTP causes the α -subunit to dissociate from the G_s complex, exposing a binding site for adenylyl cyclase on the α -subunit.

The α -subunit binds to and activates the cyclase to produce many molecules of cAMP.

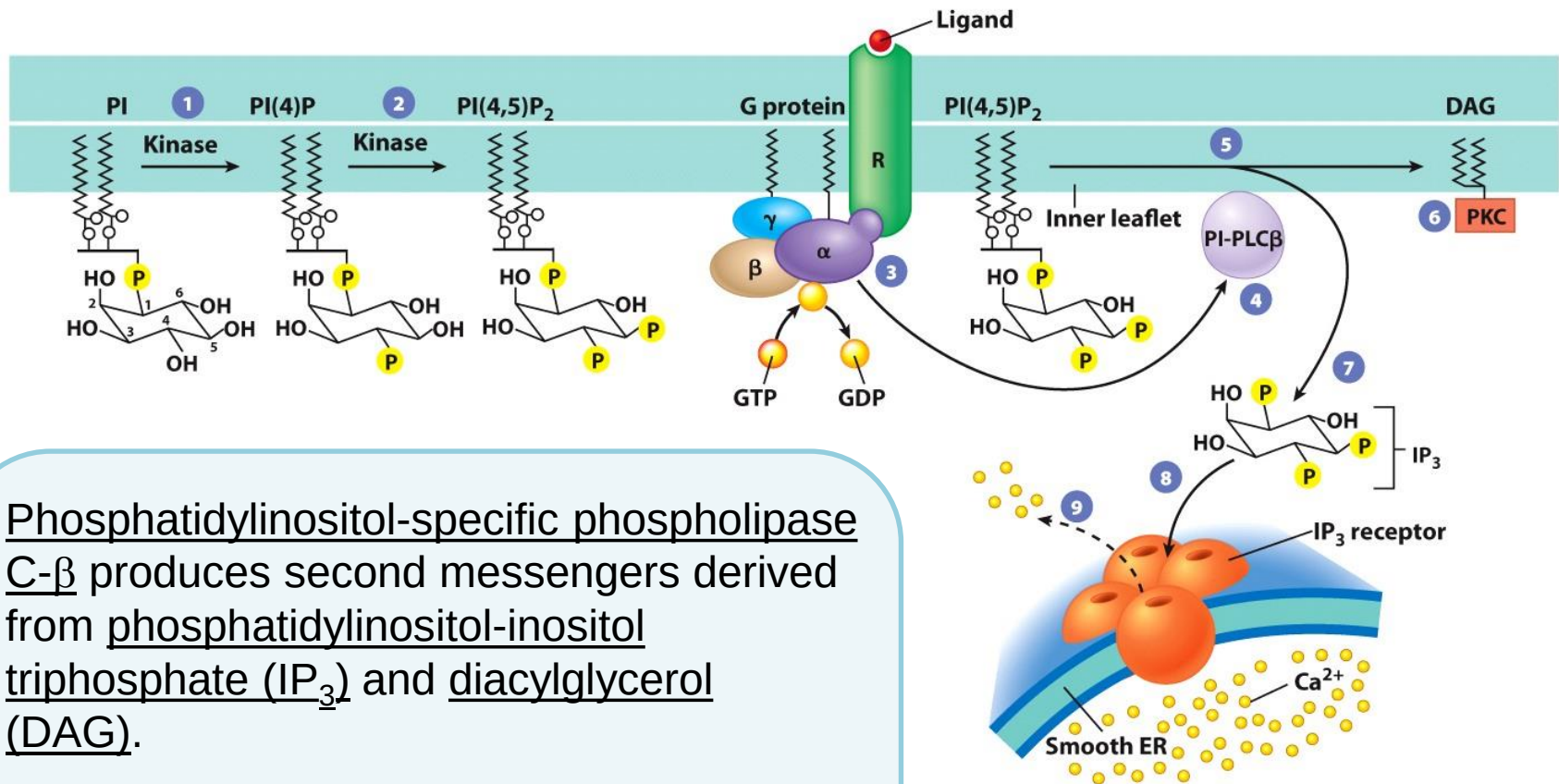
Hydrolysis of the GTP by the α -subunit returns the subunit to its original conformation, causing it to dissociate from the cyclase (which becomes inactive) and reassociate with the $\beta\gamma$ complex.

The activation of the cyclase is repeated until the dissociation of transmitter returns the receptor to its original conformation.



15.4 | Second Messengers

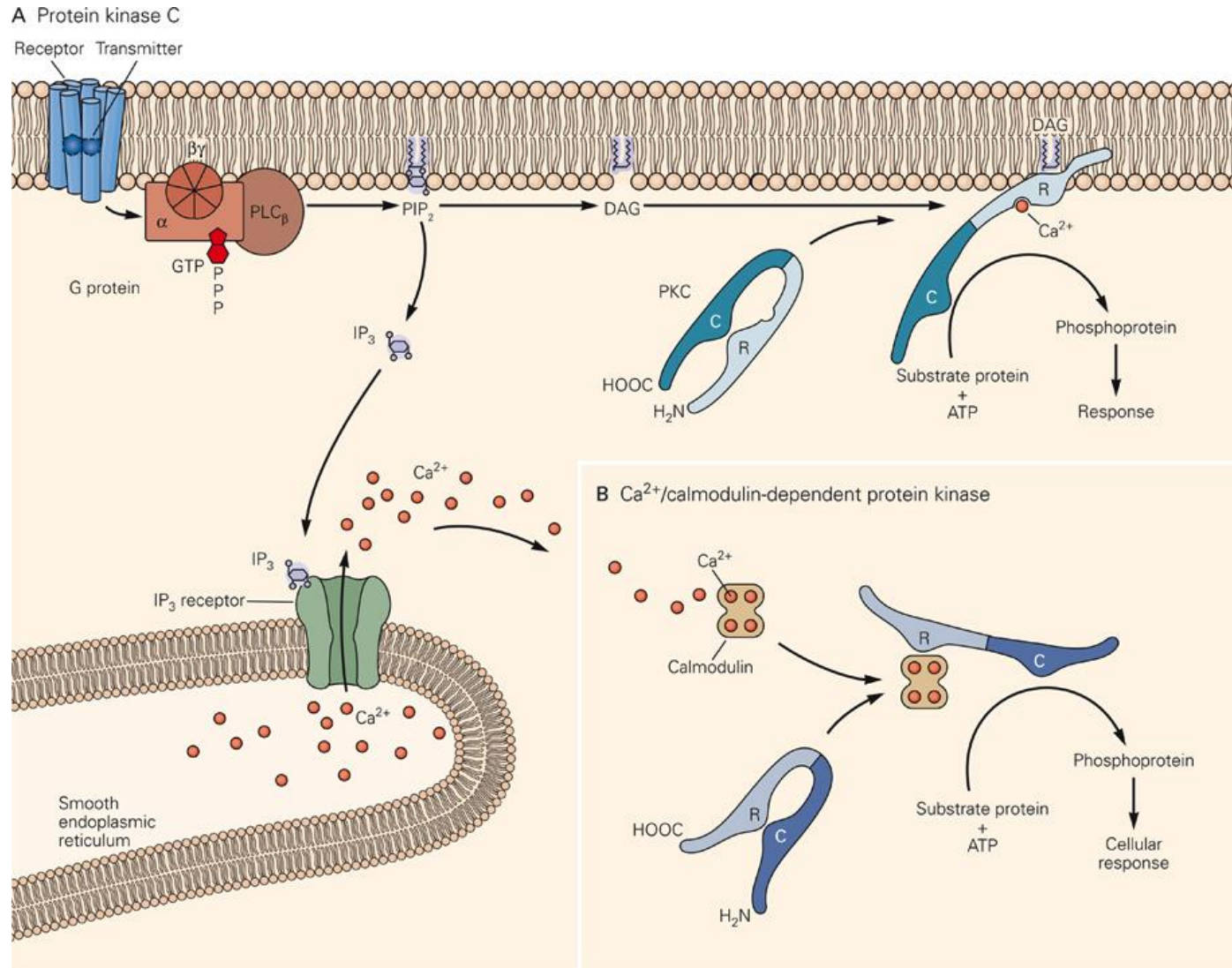
Phosphatidylinositol-Derived Second Messengers



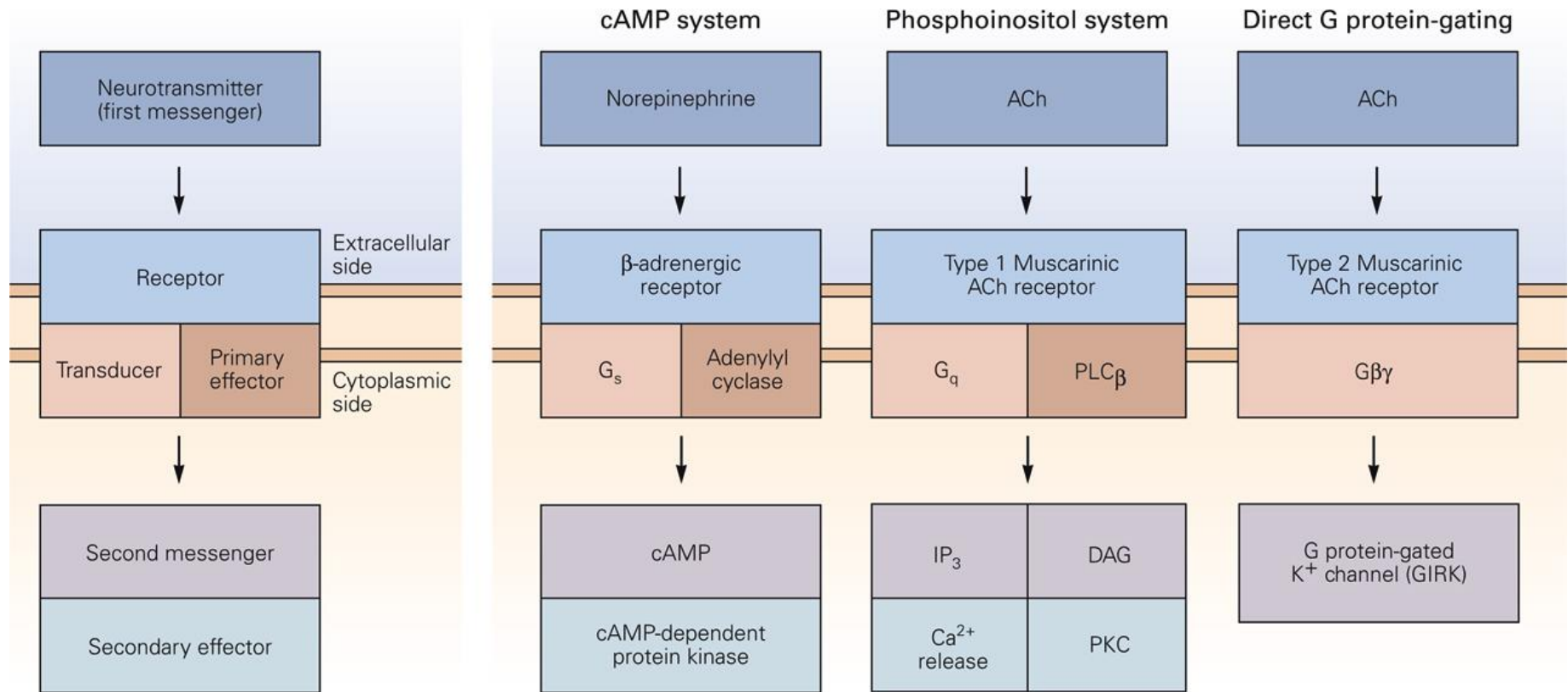
Phosphatidylinositol-specific phospholipase C-β produces second messengers derived from phosphatidylinositol-inositol triphosphate (IP₃) and diacylglycerol (DAG).

DAG activates protein kinase C, which phosphorylates serine and threonine residues on target proteins.

Hydrolysis of phospholipids activates three major second-messenger cascades



Synaptic second-messenger systems follow a common sequence

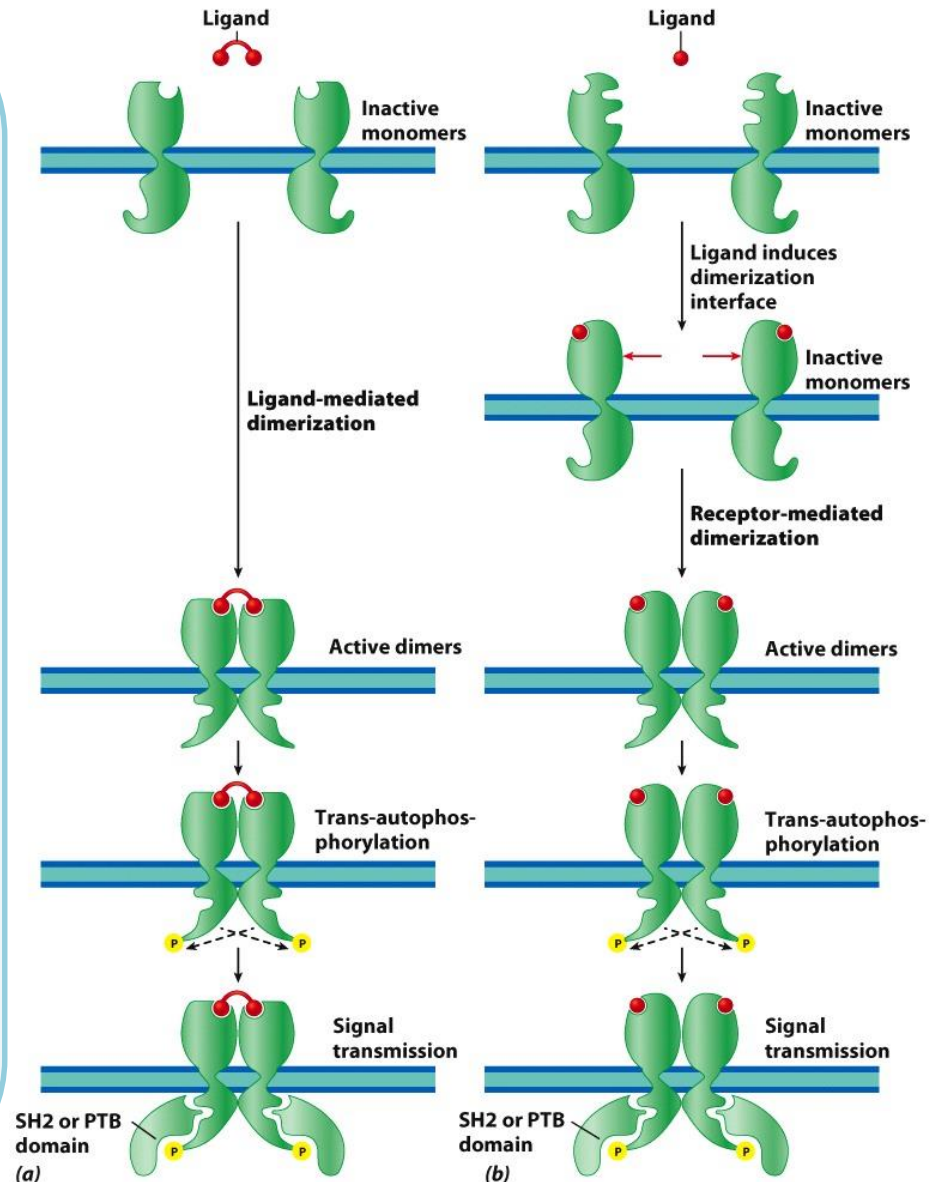


15.8 | Protein-Tyrosine Phosphorylation as a Mechanism for Signal Transduction

Ligand binding results in dimerization of the extracellular ligand-binding domains of a pair of receptors.

Two mechanisms for receptor dimerization: ligand-mediated dimerization (e.g. PDGF) and receptor-mediated dimerization (e.g., EGF).

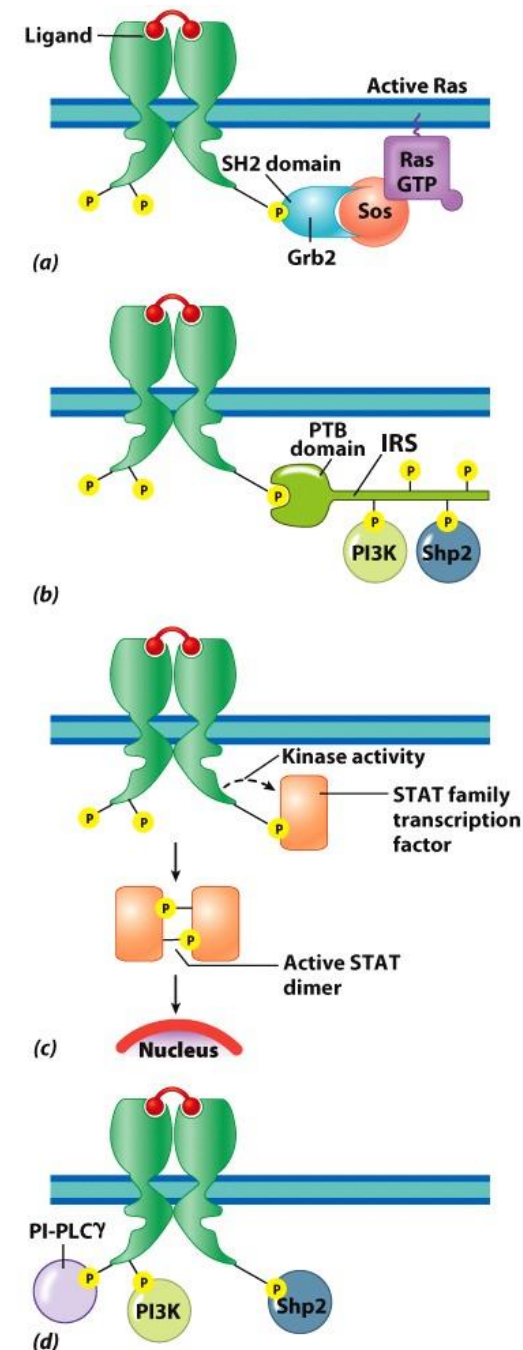
For most RTKs, dimerization brings two kinase domains in close contact for trans-autophosphorylation; the protein kinase activity of one receptor phosphorylates tyrosine residues in the cytoplasmic domain of the other receptor.



15.8 | Protein-Tyrosine Phosphorylation Activates Downstream Signaling Pathways

SH2 and PTB domain proteins include:

- 1) Adaptor proteins that bind other proteins.
- 2) Docking proteins that supply receptors with other tyrosine phosphorylation sites.
- 3) Signaling enzymes (kinases) that lead to changes in cell.
- 4) Transcription factors



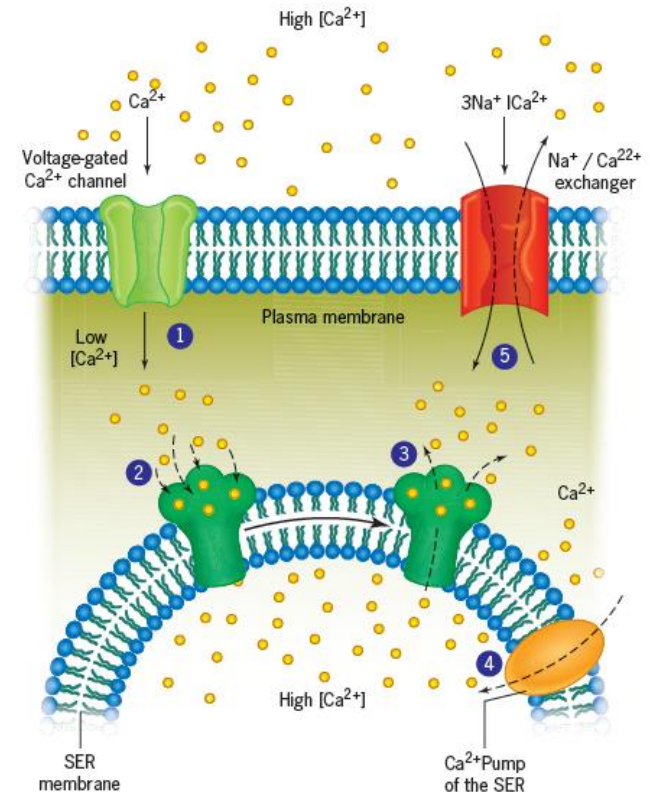
15.12 | The Role of Calcium as an Intracellular Messenger

Visualizing Cytoplasmic Ca^{2+} Concentration in Living Cells

The two main types of Ca^{2+} ion channels in the ER membrane are IP3 receptors and ryanodine receptors (RyRs), which bind the toxic plant alkaloid ryanodine.

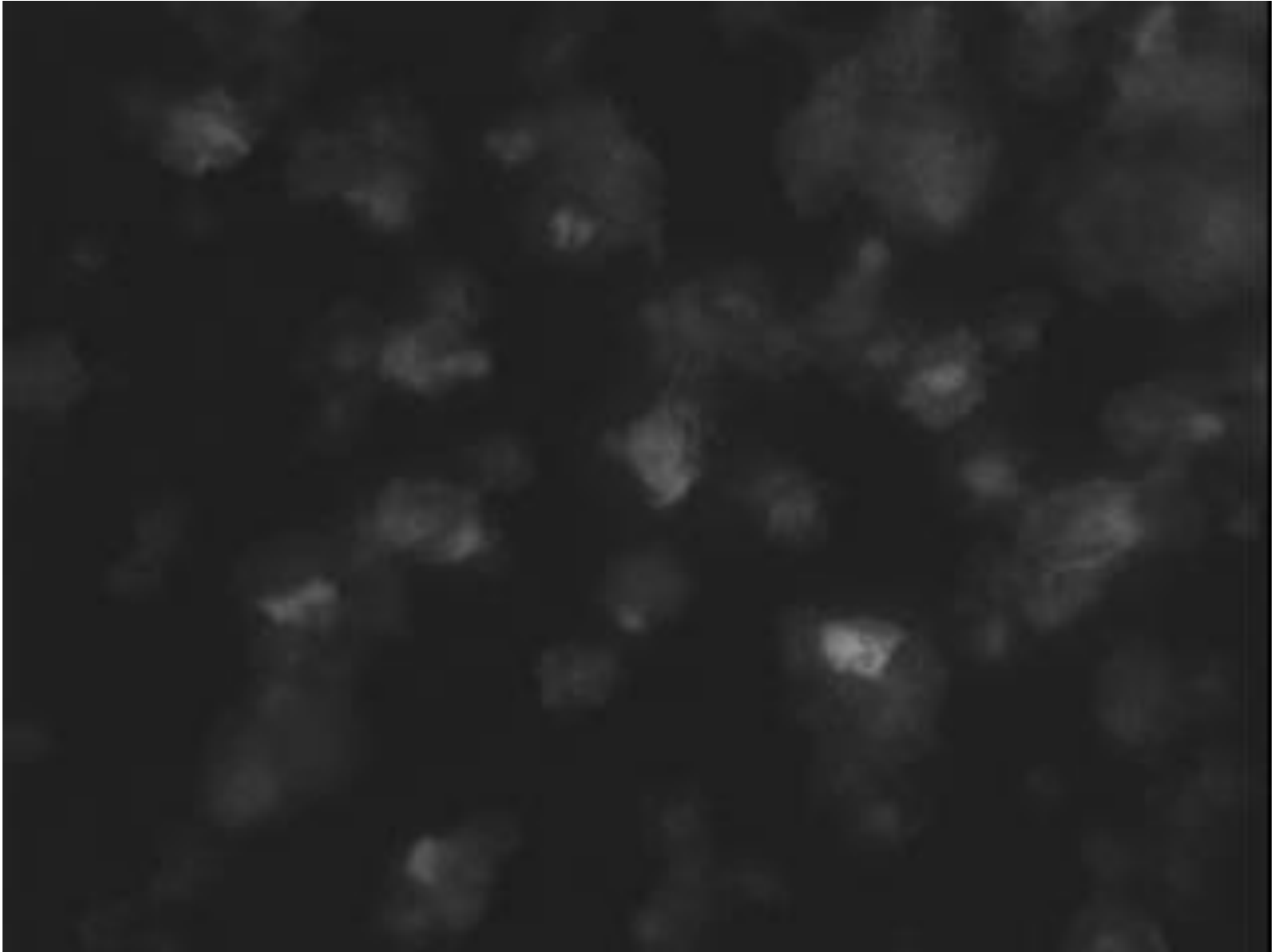
RyRs are found primarily in excitable cells like cardiac and skeletal muscle cells, where they mediate the rise in Ca^{2+} levels following the arrival of an action potential.

In calcium-induced calcium release (CICR), influx of calcium through open channels in the plasma membrane induces the opening of ryanodine receptors in the ER, causing the release of Ca^{2+} into the cytosol.

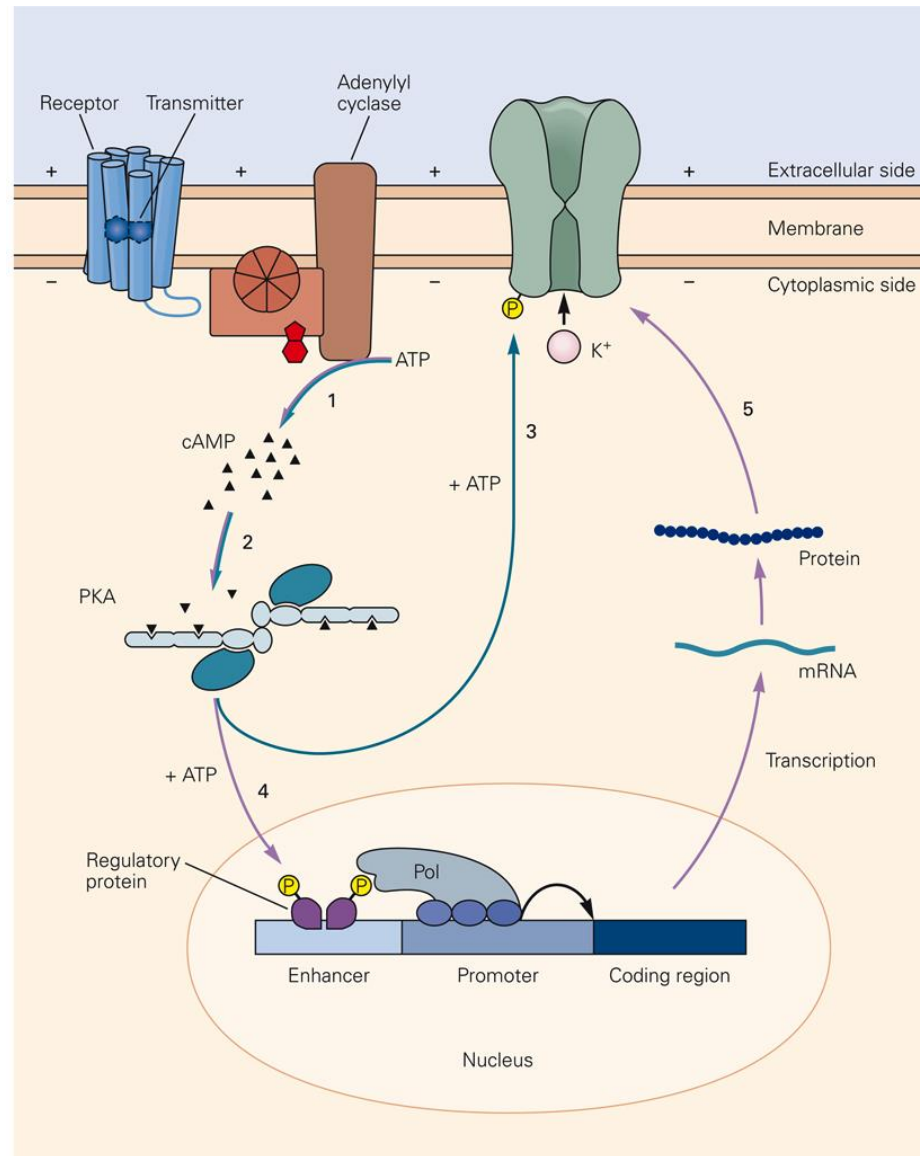


Calcium-induced calcium release: This cycle is repeated after each heart beat.

Calcium waves in platelets



A single neurotransmitter can have either short-term or long-term effects on an ion channel



Summary

G-protein coupled receptors (GPCRs), which contain seven transmembrane α helices and activate GTP-binding proteins.

Second messengers enable cells to mount a large-scale, coordinated response following stimulation by a single extracellular ligand

cAMP, Ca²⁺, phosphoinositides, inositol trisphosphate, diacylglycerol, cGMP, and nitric oxide.

Receptor protein-tyrosine kinases (RTKs), which dimerize and activate their cytoplasmic protein-kinase domain to phosphorylate specific tyrosine residues of cytoplasmic substrate proteins.

